

Impact of pharmacist-conducted home visits on the outcomes of lipid-lowering drug therapy

G. M. Peterson* PhD MBA FSHP FACPP FAIPM, K. D. Fitzmaurice* BPharm, M. Naunton* BPharm(Hons), J. H. Vial* MD FRACP, K. Stewart† BPharm(Hons) PhD and H. Krum‡ MBBS PhD FRACP

*Schools of Pharmacy and Medicine, University of Tasmania, Hobart, Tasmania, †Victorian College of Pharmacy, Monash University, Melbourne, Victoria, and ‡Department of Epidemiology and Preventive Medicine, Monash University, Prahran, Victoria, Australia

SUMMARY

Objective: To evaluate a pharmacist-conducted educational and monitoring programme, designed to promote dietary and lifestyle modification and compliance with lipid-lowering drug therapy, for patients with dyslipidaemia.

Methods: This was a prospective, randomized, controlled study. The participants were 94 adults, with 81 completing the study (intervention group: 39; control group: 42), with a cardiovascular-related diagnosis and discharged from hospital, between April and October 2001, on lipid-lowering drug therapy. Patients in the intervention group were visited at home monthly by a pharmacist, who educated the patients on the goals of lipid-lowering treatment and the importance of lifestyle issues in dyslipidaemia and compliance with therapy, assessed patients for drug-related problems, and measured total blood cholesterol levels using point-of-care testing. Patients in the control group received standard medical care. The main outcome measure was total blood cholesterol levels after 6 months, and an evaluation of patient and general practitioner satisfaction with the programme.

Results: There was no significant difference in baseline total blood cholesterol levels between the two groups. The reduction over the course of the study in cholesterol levels within the intervention group was statistically significant (4.9 ± 0.7 to 4.4 ± 0.6 , $P < 0.005$), whereas there was no

change within the control group ($P = 0.26$). At follow-up, 44% of the intervention group patients and 24% of the control group patients had cholesterol levels below 4.0 mmol/L ($P = 0.06$). The reduction in total cholesterol in the intervention group should translate to an expected 21% reduction in cardiovascular mortality risk and a 16% reduction in total mortality risk – more than twice the risk reduction achieved in the control group. In addition, the programme was very well received by the patients and their general practitioners, by satisfaction questionnaire.

Conclusion: A pharmacist-conducted educational and monitoring intervention improved the outcomes of lipid-lowering drug therapy.

Keywords: cholesterol, compliance, education, lipid-lowering drug therapy, monitoring, outcomes

INTRODUCTION

There is extensive epidemiological evidence that relates elevation of blood total cholesterol levels to increased coronary heart disease (CHD) risk (1–4). There is remarkable consistency among the large cohort studies, and CHD risk increases by 2–3% for each 1% increase in total cholesterol concentration.

In addition, many studies have now clearly demonstrated the benefits of cholesterol lowering in patients with and without established CHD, including those with average cholesterol levels for westernized societies (2, 3, 5–7).

However, the translation of the clinical trial benefits of statin therapy in CHD into practice has not been easy. Underdiagnosis, underuse of statin

Received 12 April 2003, Accepted 11 November 2003

Correspondence: Gregory M. Peterson, Tasmanian School of Pharmacy, Faculty of Health Science, University of Tasmania, PO Box 252-26, Hobart, Tasmania, Australia. Tel.: +61 3 62262197; fax: +61 3 62262870; e-mail: g.peterson@utas.edu.au

drugs and inadequate control of dyslipidaemia appear to be common worldwide (8–12). Only about 30% of patients with established CHD and raised serum lipids, and fewer than 10% of individuals eligible for primary prevention receive lipid-lowering therapy. Target total cholesterol concentrations are achieved in fewer than 50% of patients who do receive such treatment (8).

The benefits of lipid-lowering therapy will only accrue if patients comply with their medication (12, 13). The discontinuation rates with lipid-lowering therapy reported in randomized clinical trials may not reflect the rates actually observed in primary care settings (13, 14). This may be a major barrier to translating into everyday practice the beneficial effects seen in clinical trials.

A study by Simons *et al.* (15) determined that in Australian general practice, 60% of patients who were newly prescribed lipid-lowering drugs discontinue them within 12 months of commencement (half of the discontinuations occur within 3 months of initiating therapy). In a subsequent larger, Australia-wide assessment of discontinuation rates in 32 384 patients who commenced a lipid-lowering drug (HMG-CoA reductase inhibitor or 'statin' in 92%), 30% had stopped taking the drug by 6–7 months (16).

The dominant reason for non-compliance with lipid-lowering therapy in Australian patients appears to be a lack of conviction that treatment is necessary or beneficial (15, 17). According to Simons (17), the problem might be at least partially solved if both doctors and pharmacists assumed increased responsibility for convincing patients about the need for and benefits of ongoing lipid-lowering treatment. The Australian 'Lipid Management Guidelines – 2001' addressed the issue of compliance and recommended that, when possible, consideration should be given to the use of ancillary measures (e.g. special clinics, telephone support, coaching) to help patients achieve and maintain target lipid levels (3).

Therefore, the aim of the present study was to perform a randomized, controlled evaluation of a pharmacist-conducted educational and monitoring programme for hypercholesterolaemic patients, designed to promote patient compliance with lipid-lowering drug therapy and dietary/lifestyle modifications.

METHODS

Patients were recruited from the Royal Hobart Hospital, a 400-bed acute care teaching hospital and the only major public hospital in the southern region of Tasmania (serving a population of approximately 230 000). Patients with established cardiovascular disease and an acute cardiovascular/cerebrovascular-related admission, and discharged from the hospital between April and October 2001 on statin therapy, were invited to participate in the study. Patients were excluded if they had dementia, lived in a domiciliary care facility or lived beyond the greater Hobart area. Data recorded for each patient included demographic information, reason for admission to hospital, smoking/alcohol intake, medical history, medication history, management during hospitalization and drug therapy on discharge from hospital.

Sample size calculations were based on predicted compliance rates. Assuming that 30% of patients prescribed lipid-lowering drugs discontinue them within 6 months (15) and given that the aim of the intervention programme was to reduce it to 10%, it was estimated that 60 patients were needed per group (at a power of 80% and $P = 0.05$). Patients who provided written, informed consent were allocated to either the intervention or control group, using a computer-generated list of random numbers. The project pharmacist visited each patient at home 6 weeks after they were discharged from hospital. During this visit, a total blood cholesterol level was determined, to establish the baseline value (unless determined within 24 h of admission to hospital, in which case this level was used), and current medication regimen was recorded. The baseline cholesterol measurement at 6 weeks was to overcome the phenomenon of reduced total cholesterol levels (by up to 50%) that may occur within 24 h following acute coronary events and surgery (18, 19).

Patients in the intervention group were educated on the goals and proven benefits of lipid-lowering drug therapy, and appropriate lifestyle modifications. Dietary and lifestyle recommendations were obtained from various sources, including the National Heart Foundation of Australia, the American Heart Association and internet websites developed by some manufacturers of statins.

During the visit, the patient was asked the following question: 'About how often would you say

you forget to take your medications?', with response options of 'once a day', 'more than once a week, but less than once a day', 'once a week', 'once a month', 'rarely', or 'never'. For the statistical analyses, the first four categories (non-compliant) and the final two categories (compliant) were combined. Patients were also assessed for any drug-related problems. These visits were repeated on a monthly basis for a period of 6 months.

There was no further contact with patients in the control group after the initial collection of baseline data, until 6 months had lapsed. At that time, their final total blood cholesterol level was measured, and the current medication regimen and self-reported compliance were recorded.

Total blood cholesterol levels were monitored from finger-prick blood samples using the Accutrend GC (Roche Diagnostics Australia Pty. Ltd, Castle Hill, NSW), a portable instrument that measures glucose and cholesterol levels within 3 min. The accuracy and reproducibility of single-assay total cholesterol measurements with the Accutrend GC have previously been confirmed in field studies (20, 21). The target for each patient was to reduce his or her total cholesterol level to below 4.0 mmol/L (3). There was no attempt to control for recent food intake, which has little effect on total cholesterol measurements (18). Copies of the blood cholesterol result and any other findings (e.g. drug-related problems) or clinical recommendations were sent to the patient's general practitioner after each home visit. A letter was sent to every patient's general practitioner, providing all cholesterol results and any clinical recommendations that were considered necessary.

A multiple-choice 'satisfaction questionnaire', to be completed anonymously, was sent to each intervention group patient after the 6-month follow-up, along with a reply-paid envelope. Included in the questionnaire were items asking patients whether they would be willing to pay, and how much, for the cholesterol monitoring service, and whether they believed the service would be best provided through home visits by pharmacists or community pharmacies. The programme was also evaluated by surveying the general practitioners exposed to the programme to assess their opinion of its usefulness. An anonymous visual-analogue scale questionnaire and reply-paid envelope were sent to

each general practitioner who had patients in the intervention group, after the 6-month follow-up.

The statistical analyses were performed using Statview® 5.01 (Abacus Concepts Inc., Berkeley, CA, USA). Non-parametric techniques were principally used to describe patient characteristics and examine differences in variables between the groups of patients (Mann-Whitney test) and within groups between study periods (Wilcoxon signed rank test). Non-parametric analyses were used because the total cholesterol levels were skewed as a result of the minimum detectable level with the Accutrend GC being 3.88 mmol/L (any readings showing as 'low' were recorded as 3.9 mmol/L for the statistical analyses). $P > 0.05$ was considered statistically significant. Responses to questions in the visual-analogue scale survey of general practitioners were measured with a ruler to the nearest integer (in millimetres, where the length of each visual-analogue scale was 100 mm).

The study was approved by the Royal Hobart Hospital Research and Ethics Committees. A personalized letter was sent to each recruited patient's general practitioner, outlining the aims and methods of the study but not indicating to which group their patient had been allocated. Project information was also sent to the local Division of General Practice and the local branch of the Royal Australian College of General Practitioners prior to the commencement of the study.

RESULTS

A total of 210 eligible patients were approached to take part in the study, with 94 providing informed consent (Table 1). Thirteen patients were subsequently lost to follow-up; six from the control group and seven from the intervention group. Reasons for patients not completing the study were: patient moved either interstate or without leaving any forward address (4), incorrect contact details were provided at enrolment and the patient could not be contacted (4), and patient withdrew consent because of work commitments or continuing illness (5).

The principal sociodemographic and clinical characteristics of the 81 patients who completed the study are presented in Table 2. The two groups were well matched with respect to key variables. There was no significant difference in the baseline total cholesterol levels between the two groups

Table 1. Details of patient recruitment into the study

Patients approached (<i>n</i> = 210)	
Patients recruited	94
Patients not recruited	116
Reason for not being recruited	
Declined to participate	37
Discharged prior to obtaining consent	60
Participating in another study	12
Statin ceased during hospital admission	2
Died during hospital admission	2
Spouse already participating in this study	1
Dementia	1
Wrong admission diagnosis recorded	1

(Mann–Whitney $U = 802.5$, $z = 0.16$, $P > 0.80$; Table 2). Patients who were newly initiated on lipid-lowering drug therapy at the start of the study tended to have higher baseline total cholesterol levels compared with those patients who had previously been on treatment (medians of 5.1 and

4.6 mmol/L, respectively; Mann–Whitney $U = 395$, $z = 2.5$, $P < 0.05$).

Although the total cholesterol levels tended to be lower in the intervention group patients at follow-up, the difference between the two groups was not statistically significant (Mann–Whitney $U = 695$, $z = 1.18$, $P = 0.24$; Table 3). However, the improvement over the course of the study in cholesterol levels within the intervention group was statistically significant (Wilcoxon $T = 8$, $z = 3.1$, $P < 0.005$), whereas it was not within the control group (Wilcoxon $T = 16$, $z = 1.3$, $P = 0.26$).

At baseline, 28% of the intervention group patients and 29% of the control group patients had blood cholesterol levels below 4.0 mmol/L ($\chi^2 = 0$, d.f. = 1, $P > 0.90$). At follow-up, 44% of the intervention group patients and 24% of the control group patients had blood cholesterol levels below 4.0 mmol/L ($\chi^2 = 3.6$, d.f. = 1, $P = 0.06$; 95% CI of the difference in the two percentages: –1% to 40%; odds ratio: 2.5 and 95% CI: 1.0–6.4).

Patient characteristics	Control (<i>n</i> = 42)	Intervention (<i>n</i> = 39)
Age (years)		
Mean $\pm$ SD	63.5 $\pm$ 12.1	65.5 $\pm$ 11.0
Gender		
Female	13 (31%)	17 (44%)
Male	29 (69%)	22 (56%)
Lives alone	8 (19%)	8 (21%)
Number of drugs (median and range)		
Admission	7 (0–12)	6 (0–12)
Discharge	8 (3–12)	8 (4–20)
Primary admission diagnosis		
Acute myocardial infarction	4 (10%)	4 (8%)
Unstable angina	17 (40%)	12 (31%)
Angioplasty/stent insertion/ coronary artery bypass surgery	11 (26%)	14 (36%)
Atrial fibrillation	3 (7%)	2 (5%)
Stroke	1 (2%)	0
Chest pain for investigation	6 (14%)	6 (15%)
Heart failure	0	2 (5%)
On statin prior to admission	31 (74%)	29 (74%)
Length of hospital stay (days)		
Median (range)	6 (1–20)	6 (1–19)
Smoking status		
Current smoker	7 (17%)	5 (13%)
Ex-smoker	26 (62%)	19 (49%)
Baseline cholesterol level determination		
In hospital on admission	8 (19%)	7 (18%)
At home after 6 weeks	34 (81%)	32 (82%)

Table 2. Patient characteristics within the intervention and control groups

Table 3. Blood cholesterol levels within the intervention and control groups at baseline and follow-up

	Control (n = 42)	Intervention (n = 39)
Baseline cholesterol level (mmol/L)		
Median (range)	4.7 (3.9–6.8)	4.9 (3.9–6.7)
Mean $\pm$ SD	4.8 $\pm$ 0.9	4.8 $\pm$ 0.7
Follow-up cholesterol level (mmol/L)		
Median (range)	4.4 (3.9–7.8)	4.2 (3.9–5.9)
Mean $\pm$ SD	4.6 $\pm$ 0.8	4.4 $\pm$ 0.6

The improvements in blood total cholesterol over the course of the study in the intervention group were evident both in patients who were newly initiated on lipid-lowering drug therapy at the start of the study (Wilcoxon $T = 1$, $z = 2.3$, $P < 0.05$) and patients who had previously been on treatment (Wilcoxon $T = 7$, $z = 2.2$, $P < 0.05$). Lipid-lowering drug therapy (either drugs or dosages) changed for <20% of the study patients over the course of the study – 18% of the intervention patients and 21% of the control patients. Dosage increases were more common in the control group (12% of patients) compared with the intervention group (5%).

Self-reported patient compliance with medication did not change over the course of the study, and total cholesterol levels were not significantly related to self-reported patient compliance either at the baseline (Mann–Whitney $U = 449.5$, $z = 0.56$, $P > 0.50$) or at follow-up (Mann–Whitney $U = 480$, $z = 0.90$, $P > 0.30$).

The programme was evaluated by surveying the general practitioners who had patients in the intervention group, after the 6-month follow-up. Completed survey forms were received from 16 general practitioners (50% response rate). General practitioners were satisfied with the intervention, and found this to be a worthwhile service that more patients could benefit from. The intervention group patients were also anonymously surveyed after the 6-month follow-up, and completed survey forms were received from 34 patients (87% response rate). The patients were extremely satisfied with the service they received, with 97% rating the quality of the service as 'excellent'. Patients also indicated that they found the regular cholesterol monitoring 'informative' (68%), 'motivating' (47%) and 'interesting' (59%). None of the patients felt that the service was 'painful', 'a waste of time', 'annoying' or 'too frequent'. Interestingly, 80% of

patients surveyed said they would be prepared to pay for this type of service (typically \$1–\$5), and 88% felt that it would be best performed at home (as opposed to in a community pharmacy).

DISCUSSION

The intervention programme proved successful in achieving improved control of dyslipidaemia. There was a significant decline in total cholesterol levels in the intervention group over the course of the study, which was not observed in the control group. The difference in cholesterol levels between the two groups at follow-up was not statistically significant, however, mainly because the intervention group patients commenced with marginally higher median levels.

The impact of the intervention on clinical outcomes can be estimated from published data on the benefits following reduction of total cholesterol. The median reduction in total cholesterol in the intervention group over the course of the study was 0.7 mmol/L (or 14.3% of the initial value). Using the data of Gould *et al.* (22), this reduction in total cholesterol should translate to an expected 21% reduction in CHD mortality risk and an expected 16% reduction in total mortality risk. The median reduction in total cholesterol in the control group was 0.3 mmol/L (or 6.4% of the initial value). This reduction in total blood cholesterol should translate to an expected 10% reduction in CHD mortality risk and an expected 7% reduction in total mortality risk. That is, if sustained, the greater decline in total cholesterol levels in the intervention group should translate to reductions in CHD and total mortality risks of more than double compared with the control group.

A limitation was imposed by the sensitivity of the Accutrend GC monitor. The minimum measurable level for total cholesterol was 3.88 mmol/L. This

clearly does not provide much scope for measuring desirable levels below 4.0 mmol/L, the presently recommended goal (3). Twenty-three patients had baseline blood cholesterol levels below the minimum measurement limit – 11 from the intervention group and 12 from the control group. At follow-up, 24 patients had cholesterol levels below the measurement limit – 14 from the intervention group and 10 from the control group. It is highly probable that the relatively high minimum measurable level for total cholesterol resulted in an understatement of the outcome of the intervention. It is also worth noting that the baseline cholesterol levels in the study patients were relatively low, which made it more difficult for the intervention programme to achieve a significant outcome.

The reasons for improvement in hypercholesterolaemia with the intervention programme can probably be traced to education about diet and exercise, and the reinforcement offered by regular cholesterol monitoring. As dyslipidaemia is typically asymptomatic, regular follow-up is necessary to reinforce motivation and ensure dietary and drug compliance (2, 3). Lipid concentrations in patients prescribed lifestyle or pharmacological interventions must be monitored periodically to determine the efficacy of the regimen, and to serve for further counselling of the patient (2). For instance, it was noticeable in this study that almost all patients initially perceived that they had a healthy, low-fat diet. It was only when cholesterol readings were found to be above 4.0 mmol/L and rising each month that patients would disclose their true eating habits, and the actual, previously underestimated, fat content became obvious.

It is likely that these factors, rather than any direct pharmacological mechanisms, were the key elements as there was little change in drug therapy and no apparent change in compliance over the course of the study. Compliance with drug therapy did not seem to be a major barrier to achieving target lipid levels in this study, contrary to the findings in other Australian and overseas studies (12–16, 23). This was not altogether surprising. All the patients in this study possessed relatively high-risk CHD profiles and were generally well motivated, having had previous major cardiac events. Other studies suggest that the presence of pre-existing CHD is a stimulus, so that compliance with lipid-lowering drug treatment is higher when

used for secondary prevention (23, 24). Another factor that might help explain the discordance between the improvement in lipid levels but lack of improvement in self-reported compliance in the intervention group is, if the intervention improved the recognition of lapses in compliance, the intervention group may have improved the accuracy of the reporting of compliance (less under-reporting of poor compliance) at follow-up masking an actual improvement in compliance. This is unlikely to have happened in the controls.

The intervention approach in this study, with its emphasis on reinforcement and the importance of the dual aspects of pharmacological and dietary/lifestyle management, was very similar to that advocated and successfully utilized by Vale *et al.* (25), whereby patients were coached to adhere to both dietary advice and the drug treatment prescribed. Coaching may be an appropriate method to reduce the treatment gap in applying evidence-based medicine to the 'real world' (25).

Although there was a significant reduction in total cholesterol levels, there was still considerable scope for further improvement in the management of dyslipidaemia. At follow-up, 56% of the intervention group patients and 76% of the control group patients had cholesterol levels above the target of 4.0 mmol/L, despite 6 months of therapy with expensive lipid-lowering drugs. Only one-third of all the patients had achieved the target value. These findings are consistent with publications from studies worldwide showing that despite the clear evidence that reductions in total and LDL cholesterol reduce CHD risk and the existence of well-defined goals for lipid levels, targets are achieved in fewer than 50% of patients who receive lipid-lowering drug treatment (8).

The premise behind the study was that pharmacists can play an important role in detecting and preventing poor compliance, educating patients about dietary and lifestyle issues and the need to continue taking lipid-lowering drug therapy, and monitoring the response to therapy. It was also important to demonstrate that the approach was welcomed by patients and their general practitioners. The intervention programme was very well received by the patients and their general practitioners. In particular, the patients expressed high levels of satisfaction with the programme. This, along with the improvement in cholesterol levels

observed, augers well for introducing the programme on a wider scale. The intervention approach of education and monitoring in this study was similar to that successfully utilized in the Study of Cardiovascular Risk Intervention by Pharmacists (26), a randomized, multi-centre (54 sites in Alberta and Saskatchewan) study of community pharmacist intervention vs. usual care in 675 patients at high risk of vascular events.

In conclusion, this study has demonstrated the useful role that pharmacists can fulfil in reducing morbidity and mortality associated with CHD in the community, by educating patients, promoting compliance and monitoring the clinical control of hypercholesterolaemia. This approach more than doubled the reduction in total cholesterol observed with standard care. It has also shown that a pharmacist-conducted educational and monitoring programme for patients with dyslipidaemia is welcomed by patients and their general practitioners.

ACKNOWLEDGEMENTS

We thank the staff of the Pharmacy and Medical Records Departments of the Royal Hobart Hospital for their assistance with the study. Funding for this project was provided by a Community Pharmacy Practice Research Grant, through the Guild/Government (Community Pharmacy) Agreement and administered by the Commonwealth Department of Health and Aged Care. The Accutrend GC monitor and test strips were kindly provided by Roche Diagnostics Australia Pty. Ltd.

REFERENCES

1. Neaton JD, Wentworth D, Multiple Risk Factor Intervention Trial Research Group (1992) Serum cholesterol, blood pressure, cigarette smoking, and death from coronary heart disease. Overall findings and differences by age for 316,099 white men. *Archives of Internal Medicine*, **152**, 56–64.
2. Gotto AM Jr, Assmann G, Carmena R *et al.* (2000) *The ILIB Lipid Handbook for Clinical Practice: Blood Lipids and Coronary Heart Disease*, 2nd edn. New York: International Lipid Information Bureau.
3. National Heart Foundation of Australia and The Cardiac Society of Australia and New Zealand. (2001) Lipid Management Guidelines – 2001. *Medical Journal of Australia*, **175**(Suppl.), 57–88.
4. Verschuren WM, Jacobs DR, Bloemberg BP *et al.* (1995) Serum total cholesterol and long-term coronary heart disease mortality in different cultures. Twenty-five-year follow-up of the seven countries study. *Journal of the American Medical Association*, **274**, 131–136.
5. Jacobson TA, Schein JR, Williamson A, Ballantyne CM (1998) Maximizing the cost-effectiveness of lipid-lowering therapy. *Archives of Internal Medicine*, **158**, 1977–1989.
6. Heart Protection Study Collaborative Group (2002) MRC/BHF Heart Protection Study of cholesterol lowering with simvastatin in 20,536 high-risk individuals: a randomized placebo-controlled trial. *Lancet*, **360**, 7–22.
7. Hamilton-Craig I (2002) The Heart Protection Study: implications for clinical practice. *Medical Journal of Australia*, **177**, 407–408.
8. Primatesta P, Poulter NR (2000) Lipid concentrations and the use of lipid lowering drugs: evidence from a national cross sectional survey. *British Medical Journal*, **321**, 1322–1325.
9. Frolkis JP, Pearce GL, Nambi V, Minor S, Sprecher DL (2002) Statins do not meet expectations for lowering low-density lipoprotein cholesterol levels when used in clinical practice. *The American Journal of Medicine*, **113**, 625–629.
10. Pearson TA, Laurora I, Chu H, Kafonek S (2000) The Lipid Treatment Assessment Project (L-TAP): a multicenter survey to evaluate the percentages of dyslipidemic patients receiving lipid-lowering therapy and achieving low-density lipoprotein cholesterol goals. *Archives of Internal Medicine*, **160**, 459–467.
11. EUROASPIRE I and II Group (2001) Clinical reality of coronary prevention guidelines: a comparison of EUROASPIRE I and II in nine countries. EUROASPIRE I and II Group. European Action on Secondary Prevention by Intervention to Reduce Events. *Lancet*, **357**, 995–1001.
12. LaRosa JH, LaRosa JC (2000) Enhancing drug compliance in lipid-lowering treatment. *Archives of Family Medicine*, **9**, 1169–1175.
13. Applegate WB (2002) Elderly patients' adherence to statin therapy. *Journal of the American Medical Association*, **288**, 495–497.
14. Andrade SE, Walker AM, Gottlieb LK *et al.* (1995) Discontinuation of antihyperlipidemic drugs – do rates reported in clinical trials reflect rates in primary care settings? *New England Journal of Medicine*, **332**, 1125–1131.
15. Simons LA, Levis G, Simons J (1996) Apparent discontinuation rates in patients prescribed lipid-lowering drugs. *Medical Journal of Australia*, **164**, 208–211.

16. Simons LA, Simons J, McManus P, Dudley J (2000) Discontinuation rates for use of statins are high. *British Medical Journal*, **321**, 1084.
17. Simons LA (1998) Hyperlipidaemia in general practice. *Current Therapeutics*, **39**, 15–24.
18. Durrington PN (1990) Biological variation in serum lipid concentrations. *Scandinavian Journal of Clinical Laboratory Investigation – Supplements*, **198**, 86–91.
19. Rosenson RS (1993) Myocardial injury – the acute phase response and lipoprotein metabolism. *Journal of the American College of Cardiology*, **22**, 933–940.
20. Gottschling HD, Reuter W, Ronquist G, Steinmetz A, Hattermer A (1995) Multicentre evaluation of a non-wipe system for the rapid determination of total cholesterol in capillary blood, Accutrend Cholesterol on Accutrend GC. *European Journal of Clinical Chemistry and Clinical Biochemistry*, **33**, 373–381.
21. del Canizo FJ, Froilan C, Moreira-Andres MN (1996) Precision and accuracy of the measurement of total cholesterol using the reflectometer Accutrend GC. Usefulness in primary care for diagnosis of hypercholesterolemia. *Atencion Primaria*, **17**, 463–466.
22. Gould AL, Rossouw JE, Santanello NC, Heyse JF, Furberg CD (1998) Cholesterol reduction yields clinical benefit: impact of statin trials. *Circulation*, **97**, 946–952.
23. Jackevicius CA, Mamdani M, Tu JV (2002) Adherence with statin therapy in elderly patients with and without acute coronary syndromes. *Journal of the American Medical Association*, **288**, 462–467.
24. Larsen J, Vaccheri A, Andersen M, Montanaro N, Bergman U (2000) Lack of adherence to lipid-lowering drug treatment. A comparison of utilization patterns in defined populations in Funen, Denmark and Bologna, Italy. *British Journal of Clinical Pharmacology*, **49**, 463–471.
25. Vale MJ, Jelinek MV, Best JD, Santamaria JD (2002) Coaching patients with coronary heart disease to achieve the target cholesterol: a method to bridge the gap between evidence-based medicine and the “real world” – randomized controlled trial. *Journal of Clinical Epidemiology*, **55**, 245–252.
26. Tsuyuki RT, Johnson JA, Teo KK *et al.* for the Study of Cardiovascular Risk Intervention by Pharmacists (SCRIP) Investigators (2002) A randomized trial of the effect of community pharmacist intervention on cholesterol risk management. The study of cardiovascular risk intervention by pharmacists (SCRIP). *Archives of Internal Medicine*, **162**, 1149–1155.